

Title: System and Method for Quantum-Entangled Data Encryption

Abstract:

A method for securing data transmission using quantum entanglement is disclosed. The system comprises a quantum key distribution (QKD) module, an entangled photon source, and a receiver unit. The method involves generating a pair of entangled photons, transmitting one photon to a receiver, and measuring the state of the local photon to determine the encryption key. This process ensures that any interception attempt disrupts the quantum state, thereby alerting the sender and receiver to the breach.

Background of the Invention:

Traditional encryption methods rely on mathematical complexity, which may be vulnerable to future quantum computing attacks. This invention addresses the need for theoretically unbreakable encryption by leveraging the principles of quantum mechanics.

Summary of the Invention:

The present invention provides a secure communication channel using quantum entanglement. The primary component is the Quantum Entanglement Generator (QEG-X1), which produces high-fidelity entangled pairs.

Claims:

1. A system for encrypted communication comprising:
 - a. An entangled photon source configured to generate photon pairs in a Bell state;
 - b. A fiber-optic transmission medium;
 - c. A polarization-sensitive detection unit.
2. The method of claim 1, wherein the encryption key is generated dynamically based on the measured polarization states.
3. A proprietary algorithm, defined as "Q-Secure Algo v1.0", which processes the raw quantum bit error rate (QBER) to filter out noise and potential eavesdropping attempts.

Description of Embodiments:

In one embodiment, the QEG-X1 is integrated into a standard server rack. The internal process involves a non-linear crystal pumped by a 775nm laser. The "Q-Secure Algo v1.0" runs on a dedicated FPGA to ensure real-time key distillation. The confidential business knowledge regarding the specific crystal doping concentration (0.5% Mg:LiNbO₃) is critical for maintaining high photon pair production rates.